



SE2020 – Web Technologies and Mobile

Lab Activity 07: Express.js

In this lab activity, students will develop a **simple RESTful backend API** using **Node.js** and **Express.js** for a **Movie Ticket Booking System**.

The API will allow users to:

- Add movie ticket bookings
- View bookings
- Update booking details
- Delete bookings

This activity focuses on **practical implementation** and **understanding REST API flow**.

Lab Activity Tasks

A cinema needs a simple system to manage movie ticket bookings.

Each booking contains:

- Booking ID
- Movie name
- Customer name
- Number of tickets

You are required to build a backend API to manage these bookings.

Activity 1: Project Setup

1. Create a new folder named movie-booking-api

2. Initialize the project as a Node.js application
3. Install **Express.js**
4. Create a file named server.js

Activity 2: Express Server Setup Configure

an Express server that:

- Creates an Express application
- Enables JSON request handling
- Runs on port **3000**

Verify that the server starts successfully.

Activity 3: Booking Data Storage

Create an **in-memory array** to store movie booking details.

Each booking record should include:

- Booking ID
- Movie name
- Customer name
- Ticket count

Activity 4: Add a New Booking

Create an API endpoint to:

- Accept booking details from the request body
- Add a new booking to the system
- Return a success message

Test this endpoint using Postman.

Activity 5: View All Bookings

Create an API endpoint to:

- Retrieve all movie bookings
- Display the data in JSON format

Activity 6: View a Booking by ID

Create an API endpoint to:

- Retrieve a booking using the booking ID
- Return an error message if the booking does not exist

Activity 7: Update a Booking

Create an API endpoint to:

- Update booking details using the booking ID
- Allow partial updates

Activity 8: Delete a Booking

Create an API endpoint to:

- Delete a booking using the booking ID
- Return a confirmation message

